CPS607 - Autonomous Mobile Robotics
Assignment 2

General Instructions

Remember to feature your group members' **full names** and **student numbers** prominently on each part of your submission. To avoid suspicions of plagiarism **clearly state any resources (people, print, electronic), the course notes, the textbook, and/or course staff that you consulted** in your submission.

Your submission must consist of:

- A short **written report**, addressing the questions below
- Your **code** implementing the required task on your robot
- A short **video** demonstrating your robot executing the desired task

In addition, each group **must give a brief in-person demonstration to the TA during lab sessions before the deadline** of the assignment. In order to receive full marks, **each member of the group must be present** for the demonstration and be able to participate in the discussion with the TA.

Answer each question completely and justify your reasoning. Your solution will be graded not only on correctness, but also on clarity. Answers that are technically correct but difficult to understand may not receive full marks. Mark values for each question are indicated in **[square brackets]**.

Assignments must be submitted online via the D2L Brightspace Course Shell by the deadline. Late submissions will not be graded. If you require an extension due to accommodations, please contact the lecturer at least three days in advance of the deadline and consult TMU's Academic Accommodation Support. This assignment will be worth 10% of your final grade for this course.

Due Date: November 5, 2026 before 10:00pm

Marking Guide

#1	_____	/ 50
#2	_____	/ 50
Total	_____	/ 100

Toronto Metropolitan University

Task Description

Design a robust autonomous behavior for a mobile robot to clear an area of a specific object.

Setup

- The robot starts in the center of a circular arena (approx. 1 m diameter) bounded by black tape (see Figure 1).
- Two objects are randomly placed around the robot within the arena:
  1. **Target White Box:** High reflectivity; must be pushed out of the arena.
  2. **Distractor Black Box:** Low reflectivity; must be ignored and remain in the arena.
- You are allowed to modify your robot chassis with additional features like 'claws' to easier push the boxes.

Requirements

- The robot must remain inside the black boundary at all times. Touching the boundary for sensing is permitted, but crossing it may result in performance deductions depending on frequency and severity.
- The robot must locate the white box, verify its color, and push it across the boundary.
- If the black box is detected or approached, the robot should reject it and continue searching for the white box.
- Perfect performance might not be required to receive full credit. Grading emphasizes design quality, reasoning, and robustness rather than flawless execution.

Recommended Sensors

You may use any suitable sensors available in the starter kit or purchased separately. Typical choices include:

- **Ultrasonic sensor:** To locate boxes from a greater distance.
- **Binary IR obstacle sensor:** To detect box color once the robot is close to a box. The sensor threshold can be tuned using the onboard potentiometer to detect white surfaces while ignoring black surfaces due to lower IR reflectivity.
- **Line-following IR sensor module:** For boundary detection. The potentiometer can be tuned to distinguish between the arena floor and the black tape boundary.

Note: It might be necessary to place the sensors in locations that differ from the ones suggested in the starter kit. It is recommended to purchase additional mini breadboards and wires to provide flexibility in sensor location and routing.

Behavior Implementation

Implement the robot behavior using a **finite state machine (FSM)**. Examples state include `SEARCH`, `APPROACH`, `IDENTIFY`, `PUSH`, `RECOVER`, `DONE`. At a minimum, the FSM should allow the robot to search for an object, approach it, and push it out of the arena. More advanced implementations may include explicit identification, recovery, and boundary-handling states.

The FSM should handle reasonable edge cases such as:

- Losing a box while approaching or pushing.
- Inability to detect a box due to off-angle positioning.
- Difficulty identifying box color due to imperfect alignment.
- Encountering the arena boundary.

You are not expected to handle every edge case perfectly. Choose a few reasonable failure modes and demonstrate how your FSM attempts to recover.

Performance Levels

- **Excellent:** Reliably identifies the white box, pushes it out of the arena, ignores the black box, handles edge cases well, and rarely or never crosses the boundary.
- **Good to very good:** Correctly identifies and pushes the white box, but may fail in some edge cases, exhibit inconsistent behavior, or occasionally leave the boundary.
- **Sufficient to satisfactory:** Reliably pushes any box out of the arena regardless of color, demonstrating a working FSM and basic sensor usage, but without reliable object discrimination or robust edge-case handling.
- **Insufficient:** Unable to reliably push any box out of the arena.

Note: If you have tuned your sensing and algorithms using your own black and white obstacles, you may use these for the assignment demonstration and video.

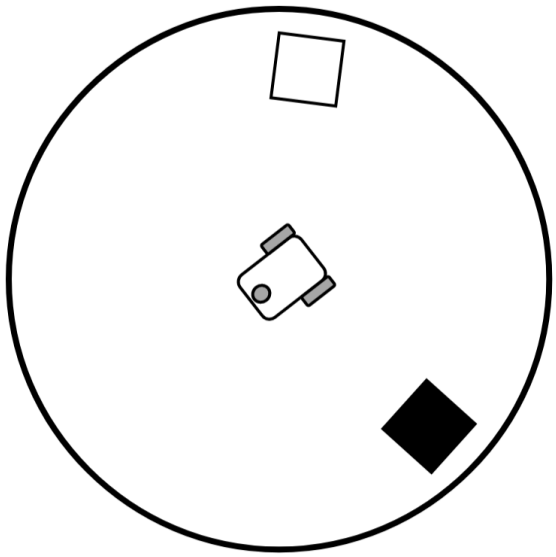


Figure 1: Task setup: The robot starts in the center of the arena. A white and a black box are randomly placed around the robot.

Question 1 - Discussion and Implementation

1. Submit a short report describing your approach and implementation. Your report should include:

   - A brief description of each sensor used, including mounting location and interaction with the environment.
   - A discussion of challenges encountered with each sensor (e.g., noise, reflections, interference, range limitations) and any steps taken to improve reliability (e.g., filtering or processing techniques).
   - A diagram of your implemented finite state machine showing all states and transitions. For each transition, clearly specify the triggering conditions based on sensor readings or timing.
   - A discussion of your FSM design, explaining the purpose of each state, the robot actions taken, and which sensors are active.
   - An evaluation of overall system performance: what worked well, what did not, and how limitations could be addressed in future improvements.

   [30]

2. Submit your full code implementation alongside the report. The code will be evaluated based on completeness and suitability for solving the task. It should demonstrate concepts discussed in class, including:

   - Non-blocking sensor reading and processing.
   - Sensor filtering where appropriate or necessary
   - Clear separation of robot behavior logic using an FSM
   - Logical state definitions, transitions, and handling of edge cases

   [20]

Question 2 - Task Demonstration and Performance

1. Demonstrate your robot to the TA during lab hours. The demonstration must include three trials with different initial configurations (e.g., box positions swapped, boxes placed at different angles or behind the robot).

   Each trial is worth **10 points**. Deductions will be made based on the performance levels described above. A video recording of the demonstration must also be submitted.

   [30]

2. During the demonstration, explain your robot's behavior and implementation to the TA. Grades will be based on your understanding of the system, ability to explain design decisions, and justification of algorithmic choices.

   [20]